

Classification:-

* This is the process of organizing data into different groups or classes based on common characteristics.

Objectives

- To facilitate comparisons i.e. to make the data comparable.
- Points out most important features of the data.
- To showcase their similarities & dissimilarities.

Types

① Geographical Classification

* Data is classified based on the location.

② Chronological Classification

* Data is classified based on the time.

③ Qualitative

* Data is classified on the basis of some attribute or quality.

* Qualitative is two types

1) Simple classification

→ Basic method of organizing data into a limited number distinct, non-overlapping categories; single criterion or attribute.

Ex:- Classifying students based on gender

└ Male
└ Female

2) Manifold Classification

→ Organizes data into more than one category simultaneously, i.e., two or more criteria.

Ex:- Male - Excellent

Male - Average

Female - Excellent

Female - Average

④ Quantitative

* Data is classified based on the measurement.

Ex:- height, weight, income, marks.

Variable (Variate)

└ Discrete Variables
└ Continuous Variables

Classification of Numerical Data

① Ordered Array or Individual Series

② Frequency Distribution

Ordered Array

* This is a simple list of all values of a variable either raw or arranged in ascending or descending order.

Ex:- 15, 18, 12, 20, 16. (Raw)

12, 15, 16, 18, 20 (Ordered)

Kinds of continuous series

① Exclusive

* In this upper limit of one class is not included in that class but in the next class.

Ex:-

0 - 10

10 - 20 $\Rightarrow$ 10

20 - 30 $\Rightarrow$ 20

} included in this classes.

② Inclusive

* In this both upper & lower limits are considered in the same class.

Ex:- 0 - 9

10 - 19

20 - 29

③ Open ended series

* One or both class intervals are open at one end i.e; no specific limit is mentioned.

Ex:- Less than 10, 10 - 20, 20 - 30, 30 and above.

Cumulative Series

* Frequencies are cumulated either from the beginning or from the end.

MEAN-

$$\text{Mean} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \quad (\text{Ungrouped Data})$$

$$\text{Mean} = \frac{\sum f \cdot x}{\sum f} \quad (\text{Grouped Data})$$

Median

$$\text{Median} = \frac{n+1}{2} \quad (\text{Ungrouped (numbers are even)})$$

$$\text{Median} = \text{middle value (numbers are odd)}$$

Grouped

$$\text{Median} = l + \left(\frac{\frac{N}{2} - m}{f} \right) \times h$$

Mode

* the value or frequency that occurs most frequently in the data.
(Ungrouped)

Grouped

$$\text{Mode} = l + \left(\frac{f - f_1}{2f - f_1 - f_2} \right) \times h$$

VARIANCE

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n} \text{ (Population)}$$

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} \text{ (Sample)}$$

STANDARD DEVIATION

$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n}} \text{ (Population)}$$

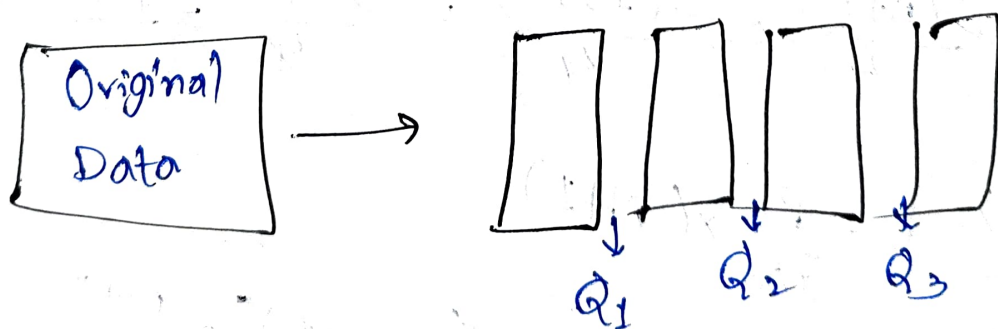
$$\sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{(n-1)}} \text{ (Sample)}$$

COEFFICIENT OF VARIATION

$$C.V = \frac{S.D(\sigma)}{\text{Mean}(\bar{x})} \times 100 \text{ (Sample)}$$

QUARTILE

"Quartiles are statistics are values that divide your data into quarters."



Ex:-

Petal Lengths

1.1

2.0

3.3

5.6

1.5

2.7

4.5

4.0

3.8

5.1

3.8

2.9

1.3

2.6

4.2

Sort the
data
→
in ascending
order

1.1

1.3

1.5

2.0 Q₁

2.6

2.7

2.9

3.3 Q₂

3.8

3.8

4.0

4.2 Q₃

4.5

5.1

5.6

$$\rightarrow Q_1 = \frac{(n+1)}{4} = \frac{16}{4} = 4 = 2.0$$

$$\rightarrow Q_2 = \frac{n+1}{2} = \frac{15+1}{2} = \frac{16}{2} = 8 = 3.3$$

$$\rightarrow Q_3 = \frac{3(n+1)}{4} = \frac{3(15+1)}{4} = \frac{3 \times 16}{4} = 12 = 4.2$$

$$\text{Rank} \Rightarrow R = \frac{P}{100} \times (n+1)$$

$$25 \text{ Percentile} \Rightarrow R = \frac{25}{100} (n+1) = \frac{n+1}{4}$$

$$50 \text{ Percentile} \Rightarrow R = \frac{50}{100} (n+1) = \frac{n+1}{2}$$

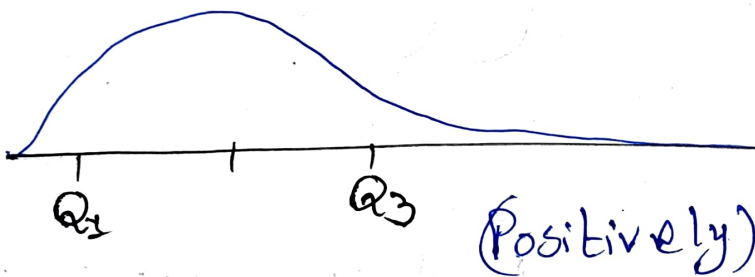
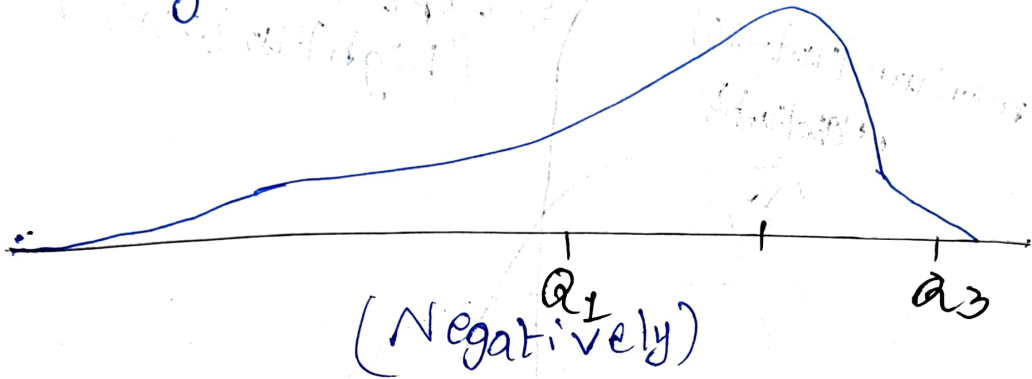
$$75 \text{ Percentile} \Rightarrow R = \frac{75}{100} (n+1) = \frac{3}{4} (n+1)$$

Empirical relationship

$$\text{Mean} - \text{Mode} = 3(\text{mean} - \text{median})$$

Skewness

* The distance between quartiles can show whether a distribution is skewed or symmetrical.



$$S_k = \frac{\text{mean} - \text{mode}}{\text{Standard deviation}}$$

Ex:
$$\frac{32 - 30.5}{10} = \frac{1.5}{10} = 0.15 \text{ (Positively skewed).}$$

$$S_k = 0 \text{ (Symmetrical)}$$

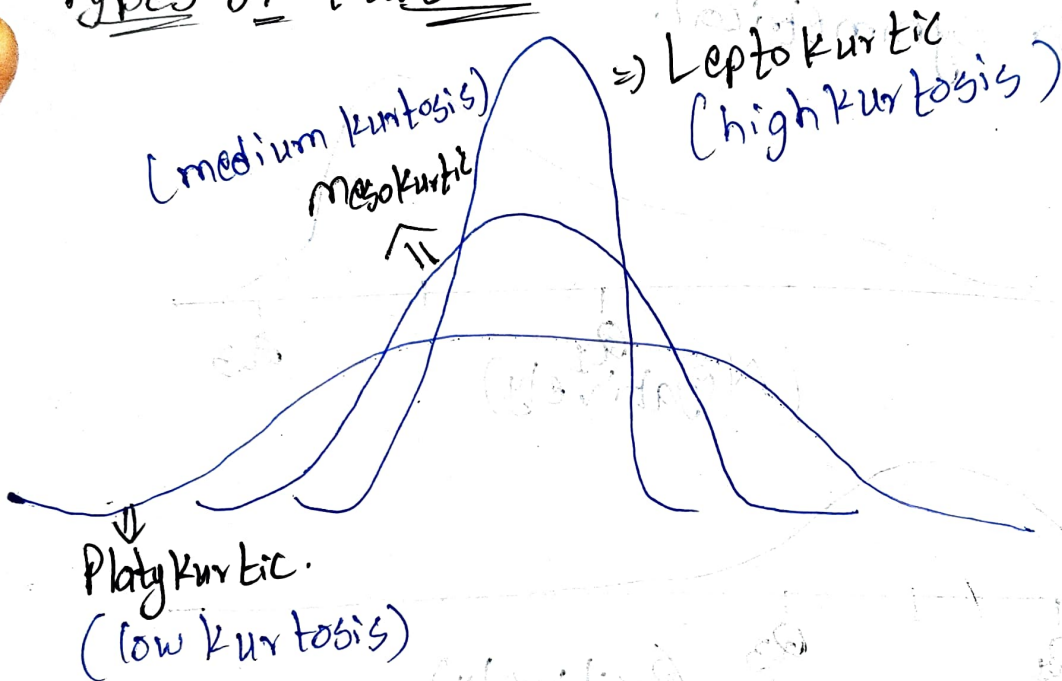
$$S_k > 0 \text{ (Positively skewed)}$$

$$S_k < 0 \text{ (Negatively skewed).}$$

Kurtosis

* It is a statistical measure that describes the shape of a probability distribution's tails in comparison to a normal distribution.

Types of kurtosis



* The moment coefficient of skewness (β_2)

$$\beta_2 = \frac{m_4}{(m_2)^2}$$

$\rightarrow \beta_2 = 3$ (mesokurtic)

$\beta_2 > 3$ (leptokurtic)

$\beta_2 < 3$ (platykurtic)

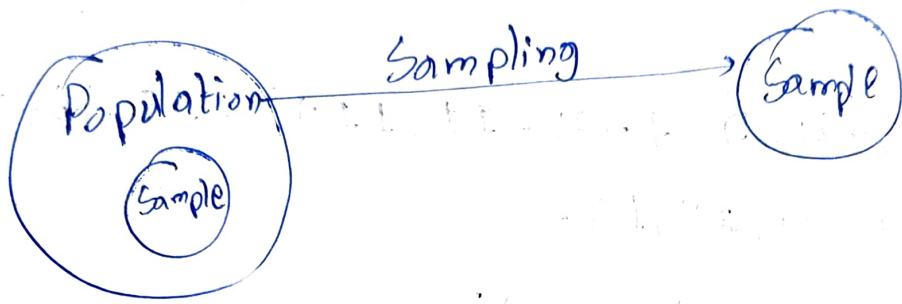
* mesokurtic $\Rightarrow$ Outliers are neither highly frequent nor highly infrequent

* Leptokurtic $\Rightarrow$ Lot of outliers (fat-tailed)

* Platykurtic $\Rightarrow$ Outliers are infrequent

Sampling

* This is used to get an idea about population based on the statistics from a subset of the population (sample).



Steps in Sampling

Step 1:- Identify and define Target Population

Step 2:- Select Sampling Frame

Step 3:- Choose sampling methods

Step 4:- Determine Sample size

Step 5:- Collect the required data.

Types

1) Probability Sampling

* In this every element of the population has an equal chance of being selected.

* Enhance equal opportunity.

Types of Probability Sampling

1) Simple Random Sampling

* Test selects randomly.

② Systematic Sampling

* The first individual is selected randomly and others are selected using a fixed 'sampling interval'.

Ex:-

3, $3+4=7$, $7+4=11$, $11+4=15$, $15+4=19$.

3, 7, 11, 15, 19.

③ Stratified Sampling

* This is used when we want to represent from all the subgroups of the population.

④ Cluster Sampling

* The population is divided into subgroups called as clusters. And a whole cluster is randomly selected.

Non-Probability Sampling

* All elements do not have equal chance of being selected.

Types

1) Convenience Sampling

Individuals are selected based on their availability.

2) Quota Sampling

* We choose items based on pre-determined characteristics of the population.

3) Judgement Sampling

* It is also known as selective sampling. It depends on the judgement of the experts.

4) Snowball Sampling

* Existing people are asked to nominate further people known to them so that the sample increases like a snowball.

Pivot Table

* This is a data analysis tool that allows you to summarize, organize & analyze large datasets.

* Used in spreadsheet applications.

Benefits:-

- Data Summarization
- Data Organization
- Data Analysis
- Flexibility.

Heat Maps

→ It is a graphical representation of data that uses colors to depict the value of a variable in a two-dimensional space.

→ Color Coding

→ Grid or Matrix

→ Data Density.

Ex- Website Analytics, Sales Data, Gene Expression, Weather Patterns.

Benefits

→ Identifies patterns quickly.

→ Improved Data Interpretation.

→ Effective Communication.

Types

→ Website Heat Maps

→ Click Maps

→ Scroll Maps

→ Mouse Tracking Heatmaps.

→ Eye-Tracking Heatmaps.

Grid Heatmaps

These are powerful visualization tool used to represent data in a tabular format.

* Identifies patterns, trends & correlations.

Pearson correlation

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Spearman's Rank Correlation

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Kendall Rank Correlation

$$\tau_B = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}$$

$$n_0 = \frac{n(n-1)}{2}$$

$$n_1 = \sum_j t_j \frac{(t_j - 1)}{2}$$

(t_j = number of values tied at j th value).

$$n_2 = \sum_k u_k \frac{(u_k - 1)}{2}$$

(u_k = number of values tied at k th value)

Point Biserial Correlation

$$r_{pb} = \frac{(\bar{y}_1 - \bar{y}_2) \cdot \sqrt{pq}}{s_y}$$

p = proportion for which nominal value is 1

q = proportion for which nominal value is 0

$\bar{y}_1$ = conditional mean when the nominal value is 1.

$\bar{y}_2$ = conditional mean when the nominal value is 0.

S_y = standard deviation of the numerical value.

ANOVA

→ Analysis of Variance (ANOVA)

* A high variance indicates that the numbers are more spread out from the mean.

* A low variance indicates they are closer to the mean.

* ANOVA is used for comparing two or more groups in data analytics.

Types

→ A one-way ANOVA (one independent variable)

→ A two-way ANOVA (two independent variables)

* ANOVA coefficient $F = \frac{MSB}{MSE}$

Most Significant Bit (MSB) = $\frac{SSB}{K-1}$

Mean Squared Error (MSE) = $\frac{SSE}{N-K}$

* SSB = Sum of squares ~~due to error~~ between

* SSE = Sum of squares due to error

* N is the total number of observations throughout K groups.

* $K-1 = df_1$ (Degrees of freedom between groups).

* $N-K = df_2$ (Degrees of freedom of errors).

* $N-1 = df = 3$ (Total degrees of freedom).

$$\rightarrow SSB = \sum r_j (\bar{x}_j - \bar{x})^2$$

$$\rightarrow SSE = \sum \sum (x - \bar{x}_j)^2$$

$$\rightarrow SST = SSB + SSE$$

* SST = Raw sum of squares - Correction factor

Types of ANOVA

1) One-way ANOVA

* Tests if there are significant differences in means among three or more groups based on one independent variable.

Factors $\Rightarrow$ One independent variable.
One dependent variable.

2) Two-way ANOVA

* Tests the effect of two independent variables on a dependent variable.

Factors:-

$\rightarrow$ Two independent variables.

$\rightarrow$ One dependent variable.